

中图法分类号: TP399 文献标识码: A 文章编号: 1006-8961(2023)06-1543-24

论文引用格式: Qiu S, Yang B H, Chen X G, Wang Y J, Xu M P, Lyu B L, Gao X R and He H G. 2023. A survey on encoding and decoding technology of non-invasive brain-computer interface. Journal of Image and Graphics, 28(06):1543-1566(邱爽, 杨帮华, 陈小刚, 王毅军, 许敏鹏, 吕宝粮, 高小榕, 何晖光. 2023. 非侵入式脑-机接口编解码技术研究进展. 中国图象图形学报, 28(06):1543-1566)[DOI:10.11834/jig.230031]

非侵入式脑-机接口编解码技术研究进展

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摘要: 脑-机接口(brain-computer interface, BCI)系统通过采集、分析大脑信号, 将其转换为输出指令, 从而跨越外周神经系统, 实现由大脑信号对外部设备的直接控制, 进而用于替代、修复、增强、补充或改善中枢神经系统的正常输出。非侵入式脑-机接口由于具有安全性以及便携性等优点, 得到了广泛关注和持续研究。研究人员对脑信号编码方法的不断探索扩展了BCI系统的应用场景和适用范围。同时, 脑信号解码方法的不断研发极大地克服了脑电信号信噪比低的缺点, 提高了系统性能, 这都为构建高性能脑-机接口系统奠定了基础。本文综述了非侵入式脑-机接口编解码技术以及系统应用的最新研究进展, 展望其未来发展前景, 以期促进BCI系统的深入研究与广泛应用。

关键词: 脑-机接口(BCI); 非侵入; 编码方法; 解码方法; 系统应用

A survey on encoding and decoding technology of non-invasive brain-computer interface

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Abstract: A brain-computer interface (BCI) establishes a direct communication pathway between the living brain and an external device in terms of brain signals-acquiring and analysis and commands-converted output. It can be used to replace,

收稿日期: 2023-01-15; 修回日期: 2023-02-28; 预印本日期: 2023-03-06

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基金项目: 国家自然科学基金项目(62020106015, U21A20388); 中国科学院B类战略性先导科技专项项目(XDB32040200)

Supported by: National Natural Sciences Foundation of China (62020106015, U21A20388); Strategic Priority Research Program of Chinese Academy of Sciences (XDB32040200)

repair, augment, supplement or improve the normal output of the central nervous system. The BCIs can be divided into invasive or noninvasive BCIs based on the placement of the acquisition electrode. An invasive BCI is linked to its records or brain neurons-relevant stimulation through surgical brain-implanted electrodes. But, it is just used for animal experiments or severe paralysis patients although invasive BCIs can be used to record a high signal-to-noise ratio-related brain signals. Non-invasive BCIs have its potentials of their credibility and portability in comparison with invasive BCIs. The electroencephalography (EEG) is commonly used for brain signal for BCIs now. The EEG-based BCI systems consist of two directions: coding methods used to generate brain signals and the decoding methods used to decode brain signals. In recent years, the growth of coding methods has extended the application scenarios and applicability of the system. Furthermore, to get high-performance BCIs, brain signal decoding methods has greatly developed and low signal-to-noise ratio of EEG signals is optimized as well. The BCI systems can be segmented into such categories of active, reactive or passive. For an active BCI, a user can consciously control mental activities of external stimuli excluded. The motor imagery based (MI-based) BCI system is an active BCI system in terms of EEG signals-within specific frequency changes, which can be balanced using non-motor output mental rehearsal of a motor action. For reactive BCI, brain activity is triggered by an external stimulus, and the user reacts to the stimulus from the external world. Most researches are focused on static-state visual evoked potential based (SSVEP-based) BCI and an event-related potential based (ERP-based) BCI. SSVEPs are brain responses that are elicited over the visual region when a user focuses on a flicker of a visual stimulus. An ERP-based BCI is usually based on a P300 component of ERP, which can be produced after the onset of the stimulus. For a passive BCI, it can provide the hidden state of the brain in the human-computer interaction process rather than temporal and humanized interaction, especially for affective BCIs and mental load BCIs. Our literature analysis is focused on BCI systems from four contexts of its MI-based, SSVEP-based, ERP-based, and affective BCI. Current situation and the application of BCI systems can be analyzed for coding and decoding technology as mentioned below: 1) MI-based BCI studies are mostly focused on the classification of EEG patterns during MI tasks from different limbs, such as the left hand, right hand, and both feet. However, small instruction sets are still challenged for the actual application requirements. Thus, fine MI tasks from the unilateral limb are developed to deal with that. We review fine MI paradigms of multiple unilateral limb contexts in relevance to its joints, directions, and tasks. Decoding methods consist of two procedures in common: feature extraction and feature classification. Feature extraction extracts task-related and recognizable features from brain signals, and feature classification uses features-extracted to clarify the intentions of users. For MI decoding technology, two-stage traditional decoding methods are first briefly introduced, e. g., common spatial pattern (CSP) and linear discriminant analysis (LDA). After that, we summarize the latest deep learning methods, which can improve the decoding accuracy, and some migration learning methods are summarized, which can alleviate its calibration data. Finally, recent applications of the MI-BCI system are introduced in related to control of the mechanical arm and stroke rehabilitation. 2) SSVEP-based BCI systems are concerned about in terms of their high information transmission rate, strong stability, and applicability. Recent researches to highlight of SSVEP-based BCI studies are reviewed literately as well. For example, some new coding strategies are implemented for instruction set or users' preference and such emerging decoding methods are used to optimize its performance of SSVEP detection. Additionally, we summarize the main application directions of SSVEP-based BCI systems, including communication, control, and state monitoring, and the latest application progress are reviewed as well. 3) For ERP-based BCIs, the signal-to-noise ratio of the ERP response is required to be relatively higher, and a single target-related rapid serial visual presentation paradigm is challenged for the detection of multiple targets. To sum up, BCI-hybrid paradigms are developed in terms of the integration of the ERP and other related paradigms. Function-based ERP decoding methods are divided into four categories of signal de-noising, feature extraction, transfer learning, and zero calibration algorithms. Current status of decoding methods and ERP-based BCI applications are summarized as well. 4) Affective BCI can be adopted to recognize and balance human emotion. The emotion and new decoding methods are evolved in, including multimodal fusion and transfer learning. Additionally, the application of affective BCI in healthcare is reviewed and analyzed as well. Finally, future research direction of non-invasive BCI are predicted further.

Key words: brain-computer interface(BCI); non-invasive; encoding method; decoding method; system application

0 引言

脑—机接口 (brain-computer interface, BCI) 是脑科学与信息科学交叉产生的新兴学科领域, 研究如何在大脑与外部设备之间建立直接的通讯和控制通道, 实现脑与设备的信息交换 (Volosyak 等, 2017)。20 世纪 70 年代, Vidal (1973) 首次提出 BCI 的概念, 并提出基于视觉诱发电位 (visual evoked potential, VEP) 的 BCI 系统框架, 开启了人类对 BCI 技术的研究。随着 BCI 技术的不断发展, BCI 技术在医疗、智能生活等领域的应用价值及其在神经科学和人脑机制研究等方面的科学价值逐渐凸显出来 (Wolpaw 等, 2002; Bermudez i Badia 等, 2013; Meng 等, 2018; Shih 等, 2012)。近十年来, BCI 逐步发展成一个研究热点, 在医疗康复、养老助残和军事科学等领域具有良好的应用前景 (张丹和李佳蔚, 2017; 曹洪涛等, 2022)。

自 2013 年起, 许多国家和地区都在战略层面对 BCI 技术进行顶层设计。美国、欧盟和日本相继开启了各自的脑计划, 我国于 2013—2014 年开始酝酿中国脑计划。2021 年, 科技创新 2030—脑科学与类脑研究重大项目发布, 标志着中国脑计划启动实施。中国脑计划包括一体两翼, 即以研究脑认知的神经原理为主体, 研发脑重大疾病诊治新技术和类脑计算与脑机智能为两翼, 以抢占全球脑科学前沿研究制高点。

BCI 主要由信号采集、信号处理和设备控制 3 部分组成 (Wolpaw 和 Wolpaw, 2017; McFarland 和 Wolpaw, 2017; Allison 等, 2007)。图 1 为典型的 BCI 系统基本结构。在通过自发或诱发方式对大脑信号编码后, 首先是信号采集阶段, 采集设备直接与大脑连接, 记录神经活动产生的生理信号, 并将信号进行放大等处理, 转化为数字信号。信号处理阶段包括特征提取和特征分类。特征提取是从脑信号中提取与任务相关且可辨识的特征, 并构造特征向量。特征分类利用特征向量对大脑信号进行区分, 实现脑电信号解码, 从而破译用户意图。设备控制阶段将特征分类后的结果转换为对外部设备的输出指令 (例如, 文字处理程序、轮椅等辅助器具或神经假体等), 以实现用户意愿 (Ortner 等, 2011; Sellers 等, 2006)。



图1 典型的脑—机接口系统

Fig. 1 Typical brain-computer interface system

按照信号采集方式的不同, BCI 可分为侵入式 (invasive) 和非侵入式 (non-invasive) 两大类。侵入式 BCI 是通过手术将电极阵列植入颅内, 直接记录或刺激大脑神经元, 以实现与外界的交流, 由于侵入式 BCI 要做手术, 因此目前仅应用在动物实验或严重瘫痪的病人上 (Joseph, 1985)。相较于侵入式 BCI, 非侵入式 BCI 的应用更加广泛, 但是通过非侵入式采集设备收集到的信号的信噪比较低。研究人员在编码实验范式以及解码方法上不断深入研究, 以提高 BCI 的应用能力 (Gao 等, 2021)。

目前, 已经有多家跨国技术公司研发并推出了多款商用脑电采集设备, 美国 EGI 公司、西班牙 NE 公司相继研发了脑电采集—电刺激调控一体机。美国杜克大学等高校与科研院所也相继研发了植入式电极 (Yin 等, 2018; Dzirasa 等, 2011a, b; Pandarinath 等, 2017)。这些硬件设备的研发加速了 BCI 领域的发展。Chen 等人 (2015) 提出滤波器组典型相关分析方法解码稳态视觉诱发信号, 构建的稳态视觉诱发电位 (steady-state visual evoked potential, SSVEP) BCI 系统突破性地实现了 151.18 ± 20.34 bit/min 的高信息传输速率, 用户用该系统可以平均每分钟输出 33 个字符。Pandarinath 等人 (2017) 提出一种高性能的皮质内 BCI 系统, 成功使 3 名瘫痪受试者通过简单的想象精准地控制电脑屏幕的光标, 并在电脑屏幕上输入了他们想说的话, 其中 1 名患者可以在 1 min 内平均输入 39 个字母, 标志着侵入式 BCI 技术前进了一大步。Anumanchipalli 等人 (2019) 提出一种新型 BCI 范式, 当参与者无声地模仿句子发音时, 该系统可以将人脑神经信号转化为语句并合成语音以实现发声交流。Edelman 等人 (2019) 利用脑电信号实现了连续的 2 维机械臂控制, 用户使用该 BCI

系统可以流畅地控制机械臂追踪光标的位置。Willett 等人(2021)利用植入式 BCI 成功解码脊椎损伤患者的手写意图并实现了每分钟输出 90 个字符,错误率只有 5.4%,再次突破性地提升了 BCI 系统的传输速度。2022 年,美国 Meta 公司通过脑电图和脑磁图以及深度神经网络模型首次实现了非侵入式 BCI 对所听到的语音的解码,其中,脑磁图的 TOP-10 解码正确率可达到 72.5%(Défossez 等,2022)。

与侵入式 BCI 相比,非侵入式 BCI 系统具有更高的安全性和广泛性,可采用的信号种类更加丰富,形成了基于脑电、脑磁、功能性近红外和功能磁共振成像的 BCI 系统。其中,基于脑电的 BCI 系统因其在使用成本、使用便携性和信号时间分辨率等方面的优势,得到了广泛关注和持续研究。在非侵入 BCI 中,面向不同领域的应用场景,研究人员对脑信号编解码方法进行了广泛且深入的探索,形成了主动式、反应式以及被动式 BCI 交互方式。主动式 BCI 指输出的控制信号反映大脑的主动性活动,不依赖于外部事件。典型代表为运动想象(motor imagery, MI)BCI。反应式 BCI 指由外部刺激引发出大脑活动,系统再根据大脑的反应而产生输出。主要包括稳态视觉诱发电位 BCI 和事件相关电位(event-related potential, ERP)BCI。被动式 BCI 不以控制为目的,在人机交互过程中为计算机提供大脑的隐性状态,从而利于计算机及时作出调整,实现人性化的交互。主要包括情感 BCI、脑力负荷 BCI 等。这些 BCI 系统在医疗康复、生活娱乐、科学研究、航空航天和军事等领域均有不同的应用。

在不同范式下,研究人员不断研发新的编码方法,以期拓宽 BCI 系统的应用场景。与此同时,由于不同范式下脑响应的机制不同,脑信号特征差异大,因此针对不同范式,BCI 解码技术也在不断创新,以实现更高的解码准确率。从传统的机器学习方法到近年来的神经网络以及迁移学习方法等都取得快速进步,展现了优秀的解码性能。BCI 编解码技术的不断发展进一步推动了 BCI 系统的实际应用。

本文围绕非侵入式 BCI,聚焦于 4 大类 BCI 系统,即运动想象 BCI 系统、稳态视觉诱发电位 BCI 系统、事件诱发电位 BCI 系统和情感 BCI 系统,分别介绍其在编码范式和信号解码方面最新的研究,以及相应的系统研发及应用进展。

1 运动想象脑—机接口研究进展

MI 表征的是一种运动意图,即想象肢体或肌肉运动,无实际动作输出(Lotze 和 Halsband, 2006)。MI-BCI 系统是一种不需要外界刺激,能够反映使用者自主运动意识的 BCI 系统,可对外部设备实现主动意愿驱动下的运动操控,并且 MI 表征的是使用者对于肢体运动的主观意图,其诱发的脑电振荡模式最接近于人体原有的运动激活模型,可促进受损脑区和神经通路的重组(Shih 等,2012)。

1.1 运动想象范式研究进展

MI-BCI 范式通常采用的是空间尺度上较好区分的运动模式,常见的运动想象部位为左手、右手、双脚和舌(Shih 等,2012; Tangermann 等,2012; Yu 等,2015; Khan 等,2020; Coyle 等,2011)。2021 年和 2022 年的世界机器人大赛锦标赛——BCI 脑控机器人大赛技能赛中 MI 赛题仍然采用的是左手、右手和双脚传统 3 类动作的 MI。然而,基于不同肢体单一动作的 MI-BCI 指令集数量有限,难以满足实际应用需求。此外,通过想象不同肢体的运动来控制外设,常常会导致运动意图与实际外设执行指令的不一致,称为认知失联(Edelman 等,2016),其会增加患者的认知负荷,严重影响 BCI 系统的使用效果。因此,设计单肢体多类别的 MI 范式并将其有效解码出来是未来 MI-BCI 发展的重要研究方向之一,已有很多研究者开始探索单关节多个动作以及单肢体多个关节的精细 MI 范式(邱爽等,2021)。

Jeong 等人(2020a)设计了单侧肢体的 MI 范式,构建了 11 种不同上肢 MI 数据库,实验任务主要包括伸臂、抓握和扭转手腕 3 个上肢运动。其中,手臂伸展包含向前、向后、向左、向右、向上和向下 6 个方向,如图 2(a)所示;抓握动作中,参与者被要求想象抓握动作以抓取 3 种日常使用的物体(卡片、球和杯子),分别对应于侧向、球形和圆柱形抓握,如图 2(b)所示;扭腕中,参与者想象手腕向左(旋前)和向右(旋后)旋转,如图 2(c)所示。之后,其团队设计了右手的伸展运动(reaching movement),包括上下左右前后 6 个方向的 MI 任务,在线抓取物品任务成功率达 60%(Jeong 等,2020b)。

Ma 等人(2020)设计了单侧肢体多关节 MI 任务,包括想象右侧腕关节运动和肘关节运动,并进行

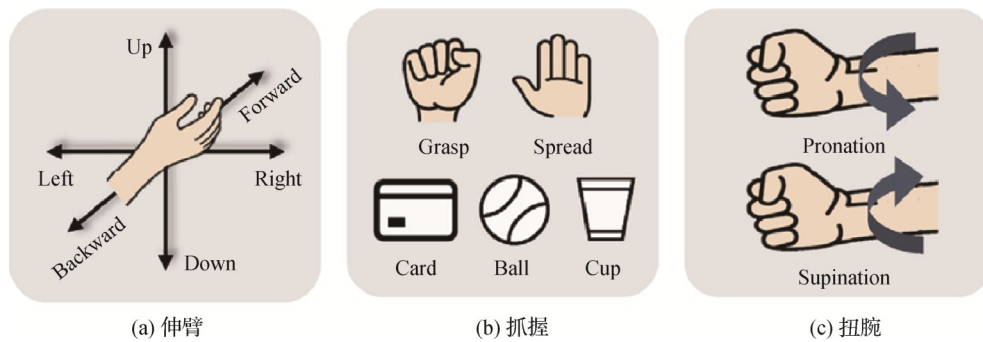


图2 11个上肢运动任务(Jeong等, 2020a)

Fig. 2 Eleven motor tasks of the same upper limb(Jeong et al. ,2020a)((a) arm-reaching; (b) hand-grasping; (c) wrist-twisting)

了数据采集、数据标准化和数据集整理,发布了首个精细运动现象数据集。2022年,该团队进一步设计了单肢体多关节5分类(包括手、手腕、手肘,肩以及静息)的运动想象任务,5分类任务的识别正确率为46.8%(Ma等,2022)。

Yang等人(2022)设计了单侧肢体4类任务,包括抓握、搬运、按键和旋转,如图3所示。受试者在想象抓握动作时需要完成打开和关闭手掌;搬运过程中,想象手拿起一个苹果,并移动一段距离;按键任务中,受试者的手和手臂需要向前伸展,并按下按钮;旋转任务则表示手腕顺时针和逆时针旋转。对4类任务设计了卷积神经网络解码模型,识别精度为47.0%。

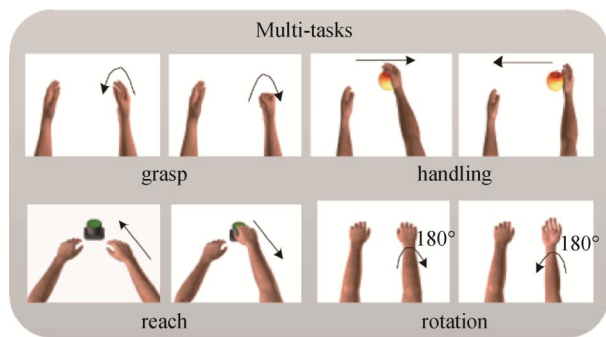


图3 单侧上肢4类任务(Yang等, 2022)

Fig. 3 Four tasks of the unilateral limb(Yang et al. , 2022)

1.2 运动想象解码方法研究进展

MI脑电解码主要分为特征提取和模式识别两个步骤。最为常用且有效的特征提取方法是共空间模式(common spatial pattern, CSP)及其衍生算法、黎曼几何以及溯源分析等(Edelman等, 2016; Ramoser等, 2000; Ang等, 2008; Barachant等, 2012)。之后,配合线性判别分析(linear discriminant analysis,

LDA)、支持向量机(support vector machine, SVM)等模式识别算法完成对不同任务的识别(Ang等, 2008, 2012)。传统脑电解码需要寻找良好的特征提取和分类算法,以及两者的最优匹配,这仍然是一个难点问题。深度学习作为一种端到端的算法很好地解决了上述问题,并在脑电信号的分析解码上有更出色的表现。基于卷积神经网络的浅层卷积神经网络(shallow CNN)、参数量较少的EEGNet、利用滤波器组共空间模式(filter bank common spatial pattern, FBCSP)特征的时间卷积—通道融合网络(channel-wise convolution with channel mixing, C2CM)模型以及镜卷积神经网络相继问世,并取得了优于传统方法的解码正确率(Schirrmeister等, 2017; Lawhern等, 2018; Sakhavi等, 2018; 罗靖等, 2021)。

Zhang等人(2021a)基于深度卷积神经网络框架,引入自适应学习方法,在大数据构建的基准模型上,使用少量被试数据参与训练,对模型部分参数重新优化,使跨被试模型性能获得提升。Mane等人(2020)提出FBCNet(filter-bank convolutional network)架构,以FBCSP结构为参考,利用滤波器组对脑电数据进行带通滤波,使用空间卷积层学习每个视图的空间判别模式。最后,使用方差层对时间信息进行聚合,在解释和泛化判别特征上具有优异表现。Li等人(2021b)提出时频压缩激活特征融合网络模型,利用小波变换提取脑电信号多频段信息并利用注意力模块对多频段信息进行压缩融合。Liu等人(2021b)提出多尺度、多任务时空频特征卷积神经网络模型,分别提取脑电信号的时空、时频特征,并利用多视图、多任务的方式优化模型特征提取能力。Ma等人(2022)提出基于时空分布注意力网络的单侧肢体MI解码模型,设计时间分布注意力网络

用于解码,首次引入注意力机制对多频带 CSP 特征进行特征权值再分配,通过滑窗保留了样本的时序信息,并用长短期记忆网络进行学习,在单侧肢体 5 分类(手、腕、肘、肩和静息)解码任务上相较已有模型进一步提升了解码正确率。Li 等人(2023)提出通过脉冲神经网络对 MI 脑电信号进行识别,只使用 1/10 的参数就可以实现与卷积神经网络相当的识别正确率。Wang 等人(2022)提出变分样本长短期记忆网络,将脑电信号分解为 6 个频带并通过样本熵筛选频带,最后使用长短期记忆网络进行分类,在 7 分类 MI 公开数据集上实现了 76.2% 的识别正确率。Liu 等人(2022b)提出基于 SincNet 的神经网络模型,将脑电信号划分为多个时段并提取 CSP 特征,再利用所提出的网络自动对信号进行滤波,最后提取稀疏表征并分类。该方法在 4 分类公开数据集的识别正确率为 74.26%。

深度学习模型往往需要大量训练数据才能获得较好性能,脑电数据的获取难度大且采集周期长,难以获得满足模型训练需求的训练数据量。因此,迁移学习等方法被引入至脑电信号解码中,以缓解模型对训练数据量需求大的问题(Wang 等, 2015)。Hong 等人(2021)融合生成对抗的思想,提出一种基于联合域自适应的方法,并使用梯度翻转层,使模型更新权重时既考虑最终的分类正确率,又保证学习的特征是源域和目标域之间共有的特征,在 4 分类运动想象任务中的识别正确率为 81.52%。Chen 等人(2022c)还将注意力机制与迁移学习融合在一起,通过多注意力层捕捉大脑中与任务相关的区域,并使用鉴别器缩小域之间的差异性,在 4 分类 MI 任务中的识别正确率为 81.48%。Chen 等人(2021b)提出基于空间注意力机制的跨被试自适应建模,该方法首先将脑电信号转换成不同频带下的时频图,然后学习源域和目标域数据之间的共有特征,之后通过最大均值差异缩小源域和目标域的特征差异,节约学习成本的同时保留目标被试的特征,在跨被试工作记忆负荷识别任务中的识别正确率为 63.6%。Chen 等人(2022f)提出对齐同类别源域与目标域样本分布、同实验组源域与目标域分布,以实现源域到目标域的领域自适应,并利用源域类别标签训练分类器,在两分类运动想象任务中的识别正确率为 82.61%。Xia 等人(2022)提出一种隐私安全的 MI 领域自适应方法,只利用源域数据的模型参数而不

使用源域脑电数据来进行迁移,在二分类 MI 中的识别正确率为 74.62%。

1.3 系统及应用研究进展

当前的 MI-BCI 系统旨在为使用者搭建一条神经通路,通过脑电信号判断患者运动意图,进而触发机械外设进行辅助工作,以实现闭环训练和主动康复。Cao 等人(2021)提出一种混合 BCI 方案,将 SSVEP 与 MI 结合应用于多维度机械臂控制,由 11 名被试完成在 3 维空间不同位置的 5 个物体的拾取任务,平均成功率达到 85%。Tang 等人(2022)针对卒中后偏瘫患者上肢运动障碍,采用基于 MI 和目标检测的混合控制方法,开发了一种可穿戴多肢机器人系统,所有受试者通过控制该系统都能在 23 s 内完成目标物体的抓取任务,其中抓取袋子任务的平均成功率最高,达到 90.67%。该系统可以通过自然的人机交互,有效帮助上肢运动障碍患者完成日常生活中的上肢任务,提高自理能力,增强对生活的信心(Tang 等, 2022)。

MI-BCI 在卒中后康复方面也有很好的应用,通过对脑卒中患者脑电信号的实时监测和反馈,促使患者主动参与训练,进而达到神经功能重塑、肢体运动功能康复的目的。Caria 等人(2020)对 30 名重度卒中后患者实施 MI 结合机械手康复治疗,通过治疗前后的功能核磁共振影像以及多种量表评估综合分析,表明 MI-BCI 加强了同侧大脑活动,增强了患侧手的感觉功能,实现了大脑对侧和同侧的体感和运动组合以及传入和传出连接相关的运动回路的重组。Ren 等人(2020)提出一种基于功能电刺激和虚拟现实的增强型 MI-BCI,降低执行 MI 任务的难度,12 名受试者的实验结果表明所提方法的分类性得到了有效提高。

总体而言,非侵入式 MI-BCI 已经在外部设备控制和脑卒中康复方面取得了很大进展,但目前仍局限于实验室环境。随着非侵入式 BCI 技术向小型化、便携及可穿戴化等方向的发展,以及神经科学和深度学习技术的进步,MI-BCI 有望实现更多更实用化的成果。

2 稳态视觉诱发电位脑-机接口研究进展

SSVEP 是受试者注视周期性的视觉刺激时诱发的视觉皮层神经元群响应(Cheng 和 Gao, 1999;

Cheng等,2002;Gao等,2003),以信息传输速率高、稳定性强以及适用性广等特点受到广泛关注,在视野损伤检测、身份识别和脑控打字界面等方面得到了广泛应用(Nakanishi等,2017;Phothisonothai,2015;Chen等,2014)。

2.1 稳态视觉诱发电位范式研究进展

设计有效的视觉刺激使用户的脑电信号可以区分不同的目标,对实现SSVEP-BCI至关重要。目前,最为广泛使用的编码范式为相位频率联合范式,即混合频率和相位编码(joint frequency-phase modula-

tion,JFPM)(Chen等,2015,2022d)方法,其由于高效的编码方式在SSVEP-BCI中占据了主流地位,研究人员近年来基于该范式开发出新型编码策略,扩大了编码指令集。Chen等人(2021c)将不同频率按照时间顺序进行排列,使用8个刺激频率,每个目标使用4种频率组合成编码序列,实现了160目标编码。Chen等人(2022d)使用12~23.9 Hz的频率范围、0.1 Hz的频率间隔以及 0.35π 的相位间隔,实现了120目标编码,如图4所示,其中,蓝色表示每个刺激编码的频率,红色代表初始相位。

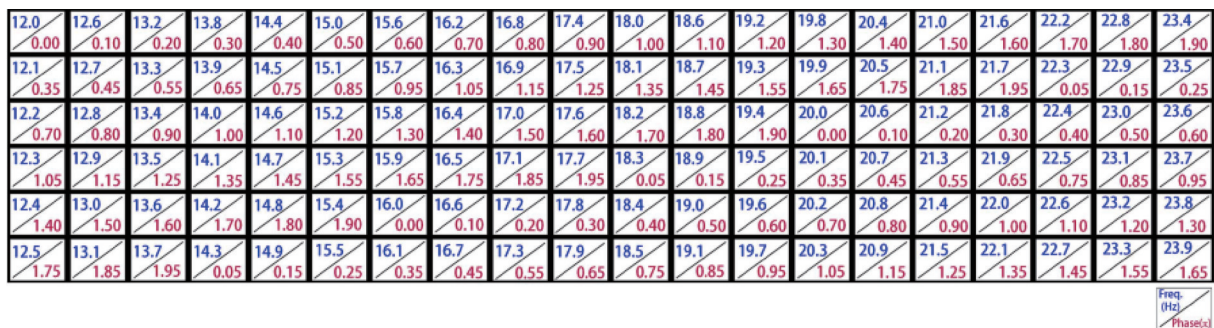


图4 120目标SSVEP型BCI系统的字符拼写器(Chen等,2022d)

Fig. 4 The 120-target SSVEP-BCI speller(Chen et al.,2022d)

传统SSVEP-BCI系统一般采用高对比度的强刺激提供视觉呈现,但易引起视觉疲劳,不利于系统的长期使用。为提高用户的舒适性,研究人员从外周视野刺激呈现的角度进行研究。Chen等人(2021a)使用4个不同频率的刺激区域,利用每个目标与4个刺激区域的位置关系,实现了13目标编码。Zhao等人(2021b)将刺激区域之间的间隔视做目标,实现了40目标的编码。Jiang等人(2022b)利用SSVEP的锁时锁相特性,使用60 Hz的无闪烁感视觉刺激结合相位编码实现了4目标的SSVEP-BCI系统,在在线测试中该系统的识别精度为87.75%,证明了无闪烁感视觉刺激实现SSVEP-BCI的可行性。

为了进一步提升编码目标数量,研究人员利用SSVEP和其他脑电范式或其他生理信号,构建了混合BCI范式。Zhou等人(2020)将SSVEP与眼电信号相结合,利用识别眨眼信号与SSVEP搭建异步12目标BCI系统。Jiang等人(2022a)开发了低频视觉刺激下(0.8~2.12 Hz)脑电响应和瞳孔响应融合范式,为缓解“BCI盲”问题提供了一个很好的解决方案。

2.2 解码方法研究进展

为了准确、高效地识别出被淹没在背景脑电噪声中的SSVEP,根据不同应用场景下可用训练数据量多寡,可将近期的研究成果分为有训练解码、迁移学习解码和零校准解码3类。

早期的有训练算法多从典型相关分析(canonical correlation analysis, CCA)算法衍生而来(Pan等,2011;Bin等,2011;Zhang等,2014,2013)。2018年,研究人员将任务相关成分分析(task-related component analysis, TRCA)引入SSVEP解码(Nakanishi等,2018),用于提取脑电中的任务相关成分,显著提升了有训练算法的性能。近些年研究者通过引入相邻频率信息(Wong等,2020)、时域信息(Zhang等,2017;Yang等,2018;Liu等,2021a)等方式,进一步提升了解码算法的性能。Liu等人(2021a)提出任务判别成分分析(task-discriminant component analysis, TDCA)算法,利用判别模型对所有频率求出共有的空域滤波器,获得了优于集成任务相关成分分析(ensemble task-related component analysis, eTRCA)算法的解码算法性能。深度学习由于强大的特征提取

能力,在有训练解码方面同样表现出了优异的性能,研究者利用基于卷积神经网络的结构分别在不同层提取空域和时域特征。Li 等人(2020b)提出卷积相关分析模型(convolutional correlation analysis, Conv-CA),解码精度超过了 eTRCA。Guney 等人(2022)提出深度卷积神经网络模型(deep neural network,

DNN),进一步提高了 SSVEP 的解码精度。Zhang 等人(2022c)考虑到脑电信号的时序特性,提出双向孪生相关分析(bidirectional Siamese correlation analysis, bi-SiamCA)模型,如图 5 所示,利用递归神经网络构建分类模型,显著提升了深度模型的解码性能。

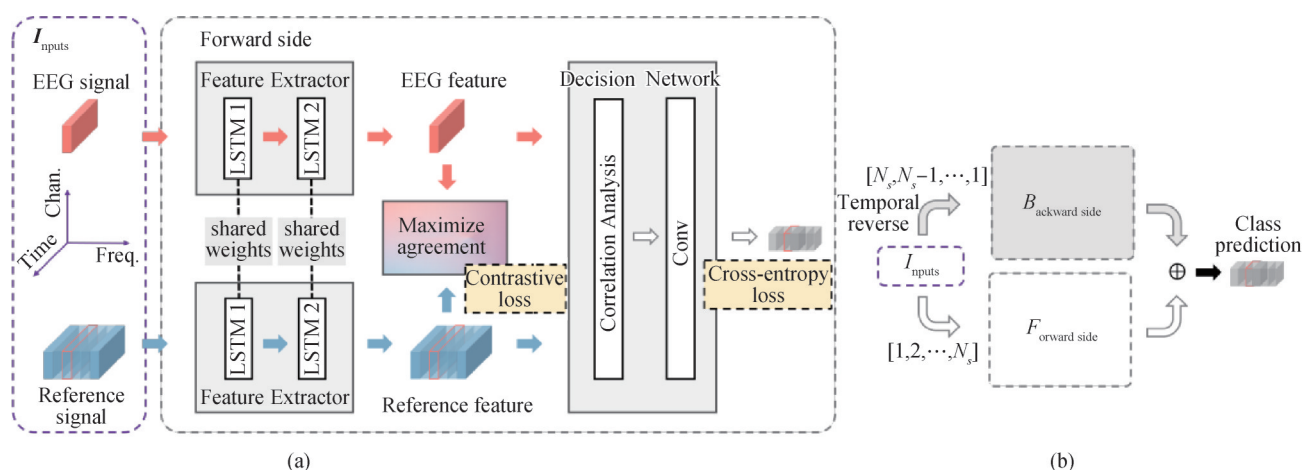


图5 高效解码模型 bi-SiamCA 原理框图(Zhang 等, 2022c)

Fig. 5 The framework of the bi-SiamCA(Zhang et al., 2022c)

长校准时间限制了 BCI 系统的推广应用。因此,研究者利用迁移学习方法构建跨天、跨设备和跨被试的解码模型,实现少校准情况下的解码。Liu 等人(2022a)使用对齐池化域适应(aligned and pool for EEG headset domain adaptation, ALPHA),从特征层面分析源域和目标域的关系,对子空间进行分解、对齐和池化操作,在被试内提升了跨设备(Liu 等, 2022e)及跨天(Liu 等, 2022e)迁移学习解码性能。Zhu 等人(2021)利用集成学习技术提升了跨天任务上的解码性能。

为了使解码方法具备直接应用在新被试上的能力,研究人员开发了零校准的解码方法。Ding 等人(2021)提出了滤波器组卷积神经网络(filter bank convolutional neural network, Fb-tCNN),在零校准情况下的解码精度显著高于传统的零训练方法 CCA。Chen 等人(2022a)将 Transformer 结构应用在 SSVEP 分类任务上,进一步提升了零校准情况下 SSVEP 的识别精度。Yan 等人(2022)通过改进跨被试空间滤波器迁移方法(cross-subject spatial filter transfer, CSSFT)的叠加平均方式,进一步提高了跨被试的 SSVEP 识别精度。此外,SSVEP 还可用于人的身份识别。Zhang 等人(2022d)通过计算 SSVEP 谱相关

矩阵得到大脑功能连接,该方法在 15 名被试数据集测试下的个体识别正确率为 98%。

2.3 系统及应用研究进展

SSVEP-BCI 主要应用于通讯、控制、状态监测 3 个方向。在通讯方面,主要用于电话拨号(Cheng 等, 2002; Hwang 等, 2013)、文字拼写(Chen 等, 2014)等;在控制方面,主要应用于家居(Gao 等, 2003)、神经假肢(Müller-Putz 和 Pfurtscheller, 2008)以及设备控制(Ortner 等, 2011; Gergondet 等, 2011; Kwak 等, 2015)等;在状态检测方面,由于 SSVEP 信号的时域、频域以及空域特征能反映大脑状态,因此用于被动式 BCI 中,实现了隐性注意程度监测(Lee 等, 2016)、注视位置监测(Chien 等, 2017)、疲劳监测(Mun 等, 2012)、视野损伤检查(Nakanishi 等, 2017)以及个体身份识别(Phothisonothai, 2015)等功能。针对实际应用需求,研究者从解码控制策略、混合 BCI 以及移动式 BCI 等方向对 SSVEP-BCI 系统进行了优化。

为提升实用性,研究者从解码控制策略方面开展了一系列研究。Shi 等人(2020)利用时空均衡动态窗算法,使渐冻症患者实现了与外界的异步中文交流。异步交流在实际应用中更自然、实用性更强,研究人员开展了异步控制方法研究。Yang 等人

(2021)提出了基于空时均衡的多窗异步检测策略,提升了 SSVEP-BCI 键盘的异步控制性能。此外,考虑到训练校准时间长、易引起疲劳的问题,可以设置在线适应策略,通过在线机器学习提升系统在少训练或无训练条件下的性能。Wong 等人(2022)提出在线自适应的 CCA(online adaptive CCA, OACCA),通过在线数据对算法模型进行不断迭代更新,获得了比动态停止策略更优的性能。

除了从控制策略方面提升实用性,研究人员也从穿戴设备角度进行了改进。为方便信号采集,提出了耳后信号采集方法。Marinou 等人(2020)利用单耳后采集 SSVEP 响应进行目标检测,验证了耳后 SSVEP-BCI 的可行性。Liang 等人(2021)提出一套快速有效的左、右视野耳后叠合延迟差的估计方法,有效提升了耳后 SSVEP-BCI 的系统性能。为加速刺激范式和控制场景切换,Chen 等人(2020)使用移动式 BCI 和增强现实技术,通过 4 目标 SSVEP-BCI 系统控制机械臂,实现了 14.21 bit/min 的信息传输速率,Chen 等人(2021b)将控制指令提升至 12 目标,实现了 67.37 bit/min 的信息传输速率。

3 事件相关电位脑—机接口研究进展

ERP 是一种特殊的脑诱发电位,通过有意地赋予刺激以特殊的心理意义,利用多个或多样的刺激引起认知过程中大脑的神经电生理改变(Donchin 和 Lindsley, 1969)。ERP 常作为 BCI 的控制信号。

3.1 诱发范式研究进展

目前常用的 ERP-BCI 编码范式有快速序列视觉呈现范式(rapid serial visual presentation, RSVP)和 P300 范式等(Lees 等, 2020; Chen 等, 2022e)。但是, P300 范式诱发的 ERP 信号信噪比低,而传统 RSVP 范式只针对单个目标的识别,只能应用于单一固定目标场景下的目标检测。基于上述局限性,有研究人员将 ERP-BCI 的不同范式组合在一起,开发混合 BCI 范式来提升系统性能(迟新一等, 2022)。Xu 等人(2020)开发了一个基于 P300 和 SSVEP 的高速混合 BCI 范式,采用时间编码和频率相位组合编码的方式分别对 SSVEP 和 P300 信号进行编码,将 BCI 指令集扩展到高达 108 个,如图 6 所示,信息传输速率最高可达 238 bit/min。

Ko 等人(2021)提出一种基于 SSVEP 和 RSVP 的

>>											
F1	F2	F3	A	B	C	J	K	L	back	Enter	shift
F4	F5	F6	D	E	F	M	N	O	space	Alt	Ctrl
F7	F8	F9	G	H	I	P	Q	R	insert	Tab	Esc
F10	F11	F12	S	T	U	1	2	3	Num	Caps	%
Fn	Win	?	V	W	X	4	5	6	PgUp	Home	^
~	'	!	Y	Z	0	7	8	9	PgDn	End	&
(	)	@	"	'	Del	-	+	=	,	.	
[	]	#	PreSc	Config	ScrLK	*	/	\	:	;	
{	}	\$	Relick	Pause	LeLick	←	↓	→	<	>	_

(a) 108 个字符由红色虚线区分为 12 个子拼写区

1	2	3	4	
12.4 / 0	12.6 / 0.35	12.8 / 0.70	13.0 / 1.05	
5	6	7	8	
13.2 / 1.40	13.4 / 1.75	13.6 / 0.10	13.8 / 0.45	
9	10	11	12	
14.0 / 0.80	14.2 / 1.15	14.4 / 1.50	14.6 / 1.85	
				No. of sub-speller
				Freq / Phase (Hz) / (π)

(b) 每个子拼写区的频率和初始相位

图 6 混合 BCI 拼写器系统(Xu 等, 2020)

Fig. 6 The hybrid BCI speller(Xu et al. , 2020)

((a)distribution of 108 characters on the screen was divided into 12 sub-spellers by the red dash lines;

(b)the selected frequency and initial phase of stimulation squares were displayed for each sub-speller)

混合 BCI 范式,在这种范式中,SSVEP 刺激用于识别用户的焦点位置,RSVP 刺激区分目标和非目标物体,提升了多目标场景中目标/非目标对象的分类性能。随着 BCI 系统在多个领域的应用推广,人们开始尝试开发更舒适的 BCI 控制系统。近几年,研究人员开展了对周围视野刺激的研究,以提升系统的舒适性和交互友好性。Zhou 等人(2021)提出一种基于视野弱刺激的新范式,采用一簇极小的圆点进行视觉刺激,并在中央视野周围选择了 16 个不同的位置进行分析,该研究实现了对极微弱脑电的编解码,同时证明了周围弱刺激的可行性,并提供了一种潜在的方法来扩大 BCI 的应用范围。Zhang 等人(2022a)提出一种具有 3 个视觉空间模式的 RSVP 刺激的新范式,该研究提供了视野不均匀性与脑电图特征之间关系的见解,并为实现 RSVP-BCI 中目标空间方向的精确定位提供可能。

针对 BCI 范式,研究人员不仅在编码方式进行改进,还在交互方式和刺激效果上进行了新的尝试。Gehrke 等人(2022)设计了一个用于评估视触觉匹配与否的范式,设置了匹配/不匹配两组实验,其中,匹

配与否指被试触摸到物体是否会收到反馈刺激,该研究可为虚拟现实视触觉感觉整合冲突的自动检测提供一种新思路。Korkmaz 等人(2022)提出一种基于3D刺激的新型P300范式,证实了3D刺激的可行性,并发现了3D刺激可获得更好的ERP响应。

3.2 解码方法研究进展

本文调研了ERP-BCI的解码算法,并根据这些方法的功能归纳为4类,即信号降噪算法、特征提取算法、数据迁移算法以及零校准算法。

脑电信号具有幅值低、信噪比低的特点,如何从各种噪声中将ERPs提取出来,是ERP-BCI研究中的一个重要问题。Yang 等人(2020)设计了一种时空均衡方法,有效提高了ERP信号的信噪比。Sosulski 等人(2021)通过将噪声与时间解耦,增强了LDA的分类性能。Yadav 等人(2022)设计了一种基于海洋捕食者算法的自适应滤波器,实现了对ERP信号的去噪。除了脑电背景噪声的滤除外,研究者通过剔除混杂在脑电中的其他生理信号(如眼电,肌电等)同样增强了ERP-BCI的性能(Ouyang 等, 2022; Liu 等, 2022f)。

在ERP信号的特征提取过程中,可以从时域、空域等多个维度挖掘潜在的信号特征。针对时域特征,Tan 等人(2021)提出一种多窗口时空特征分析的卷积网络,以缓解RSVP中单次脑电数据的时间变异性。Rocco 等人(2021)提出一种可以从单次中提取ERP成分的方法,实现了试次数量的扩增。Zhou 等人(2021)开发了一种多类DCPM(discriminative canonical pattern matching)分类器(Multi-DCPM),实现了对多模式ERP信号的检测和识别。针对空域特征,Xiao 等人(2021)提出一种DCPM空间滤波算法,显著增强了aVEP的信噪比,并在同年证明了DCPM在ERP模式识别上的鲁棒性。Tang 等人(2021)将矩阵映射到不同空间,也实现了ERP空间特征的提取。针对时一空混合特征,Li 等人(2021a)同时对数据进行空间滤波和时间加权,实现了在小样本条件下的性能提升。Xiao 等人(2021)在对P300拼写器的研究中,考虑到ERP模式随时间变化的特点,提出了多窗口DCPM(multi-window DCPM, MWDCPM),构造了一系列与时间相关的DSP滤波器来微调空间ERP特征的提取。崔玉洁等人(2022)分别在时域和空域采用主成分分析和共空

间模式对脑电信号进行特征提取。

传统BCI分类器往往依靠用户的个人校准数据作为训练模板,而脑电信号的变异性导致用户个体的校准数据在跨被试、跨时间情况下变得不再适用,不得不重新采集校准数据,导致ERP-BCI系统在使用时训练过程烦琐、冗长,限制了实际应用。针对这一问题,研究者提出了多种基于迁移学习的算法,实现了训练样本的跨时间、跨被试迁移,有助于降低ERP-BCI的训练负担,增强其实用性和交互友好性。Wei 等人(2020)提出一种多源条件对抗域自适应与相关度量学习框架,利用来自其他被试的数据,以减少来自新被试的训练数据需求。Li 等人(2022b)提出一种基于3维CNN(3D-CNN)的ERP动态解码模型,利用3维卷积核捕捉空间特征在多个时间点的动态特征来提高模型在不同被试间的泛化性能。Abibullaev 等人(2022)提出一种模型选择方法,通过使用留一被试法,利用跨被试交叉验证选择CNN超参数。

目前ERP-BCI解码算法在每个用户使用BCI系统前需要进行校准,采集用户带标签的脑电数据用来训练用户特异的脑电解码模型。数据迁移算法虽然能够利用已有被试数据来降低对当前被试数据的依赖,但仍然需要采集当前被试数据。零校准是让新被试可以使用大量其他被试数据提前训练好的模型直接工作,不用再采集被试自己数据训练自己特定的分类器,不仅节省了使用前的等待时间,也提高了BCI模型的通用性。Lee 等人(2020)使用55名受试者的脑电数据联合训练一个EEG-Net模型,该模型可直接用于新被试的脑电解码,且零校准模型的性能与被试依赖模型的性能无显著性差异。Wei 等人(2022)提出一种基于元学习的原型匹配网络,通过在大量已有被试的数据上学习到P300目标与非目标的原型,测试被试的脑电数据在网络的特征空间中直接与原型计算相似度即可实现分类;Li 等人(2022d)提出一种脑电时频融合模型TFF-Former(temporal-frequency fusion transformer),利用Transformer表征能力强的特点,提取被试的时频共性特征,提升零校准场景下的模型性能,实验结果表明,所提出模型的零校准性能与有校准模型利用4个block校准数据训练的性能相当。

3.3 系统及应用研究进展

P300拼写器是ERP-BCI最经典的应用,利用

P300的潜伏期和振幅对注意力和认知程度的反应,可以实现控制命令的选择和字符的拼写。Townsend和Platsko(2016)基于P300开发的异步拼写器通过动态调整刺激时间,实现了最高120 bit/min的信息传输速率。在P300拼写器的基础上,P300-BCI系统还应用于智能手机的电子邮件发送(Velasco-Álvarez等,2021b)、无人机控制(Kim等,2021)、导航框架内决策(Vařeka和Ladouce,2021)和智能病房协作系统(Cai和Pan,2022)等。Chang等人(2020)根据P300提供的隐藏信息开发了隐藏信息测试方法用于测谎。此外,P300因其特征明显、不过于依赖用户训练等优势,也用于游戏设计(何峰等,2022)。

ERP-BCI不仅可以帮助健康人群实现外设控制,也可以为行动不便的患者提供辅助和支持,例如帮助患者进行轮椅或鼠标控制、帮助肌萎缩侧索硬化症患者控制BCI家庭自动化系统(Velasco-Álvarez等,2022)、帮助残疾人对假肢装置进行意志控制(Hasan等,2021)等。除此之外,ERP-BCI在研究感官信息处理、心理病理状况以及高阶认知等方面也很有应用价值(Habelt等,2021)。

在意识状态检测方面, Ji等人(2022)通过N300延迟诊断学习障碍儿童;Kool等人(2022)通过P3a、P3b和失匹配负波测量精神障碍的注意缺陷;Xiao等人(2022)根据定向听觉刺激诱发P300和失匹配负波诊断意识障碍患者;Pan等人(2022)设计了一种利用P300电位与情绪模式的混合BCI系统应用于检测意识障碍患者的意识。

在治疗情绪相关疾病方面,P300、晚正电位和反馈相关负波等用于监测情绪调节障碍,减少自闭症谱系障碍患者的症状(Eldeeb等,2021),通过测量患者识别自我和他人面孔的P3差异,表明自闭症谱系障碍患者对自己面孔的注意力偏向受损(Cygan等,2022)。此外,奖赏积极性和晚正电位可用于抑郁症的诊断(Klawohn等,2021)。

在帮助患者功能康复方面,通过N1和P2分析视听感知,补偿人工耳蜗输入,进而改善患者的听觉康复(Layer等,2022;Radecke等,2022);通过对脑震荡后运动员P3B的测量,提供运动员康复的心理情感、认知和神经生理学特征(Sicard等,2021)。

在高级认知检测方面,体感P300用于测量冥想者的注意力资源(Bokk和Forster,2022),晚正电位用于研究工作记忆负荷对情绪调节的影响(Adamc-

zyk等,2022),N/P150、N250和N400等用于探索音韵学在视觉单词处理中的作用(Shetty等,2022),P300和失匹配负波用于研究音频流向用户提供可理解的内容是否可以帮助用户更好地关注于目标音频流,从而更好地关注目标刺激且忽略非目标刺激(Velasco-Álvarez等,2021a)。

4 情感脑-机接口研究进展

4.1 情绪范式研究进展

目前,情绪诱发范式采用的素材大致可分为图像、文本、音频、视频以及游戏等。图像诱发范式中,刺激素材包括国际情感图像系统(international affective picture system, IAPS)(Lang等,1997)、中国情感图像系统(Chinese affective picture system, CAPS)(白露等,2005)和中国面部情感图像系统(Chinese facial affective picture system, CFAPS)(龚栩等,2011)等。文本诱发范式则是让被试朗读情感词汇、朗读诱发不同情绪的文章(Guo等,2021)、看图叙述故事(Valstar等,2013)、描述正向或负向情绪的事件(Alghowinem,2013)等。音频实验范式是指利用音乐作为辅助的刺激手段,要求被试在聆听音乐的同时回忆悲伤的经历(Steidtmann等,2010)。而视频片段诱发范式同时结合了视觉和听觉,相较于图像、文本与音频,诱发强度更高,是目前最为常用的实验范式。许多情绪脑电数据库采用了视频片段诱发的范式。例如,在DEAP(A database for emotion analysis using physiological signals)数据库中(Koelstra等,2012),32名受试者分别观看了40段1 min的音乐视频。SEED(SJTU emotion EEG dataset)系列数据集包含了多个不同被试数量的子集,分别诱发被试的3类、4类与5类离散情绪。其中3类情绪也根据被试国籍区分出了中国、法国与德国3个国家被试的数据子集(Liu等,2022d)。DREAMER数据集(Katsigiannis和Ramzan,2018)使用了18个电影片段来诱发23名被试的情绪,并采用了连续情感标签。

研究者对视频诱发素材和数据标签划分等进行创新。Luo等人(2022)将油画作为一种新型的图像诱发素材,如图7所示,相较于人像或风景照片,油画获取更为简便,并包含丰富情绪情景。Hu等人(2022)基于视频片段诱发范式,采集了80位被试9类情绪类别的THU-EP(Tsinghua University emo-

tional profile dataset)数据集,类别精细度大幅超越了传统正面、中性、负面的3分类,或效价、唤醒度维度上的2分类。



图7 油画作为情绪诱发素材的数据集(Luo等,2022)

Fig. 7 A emotion dataset with oil paintings stimulus
(Luo et al., 2022)

4.2 解码方法研究进展

研究者提出了多种新型脑电情感解码模型来提升情感BCI的分类性能。Li等人(2022e)提出一种新型的层级化时空特征提取模型,分别提取区域和全局的时空特征,提高了模型的分类能力。Li等人(2022a)提出一种基于多目标进化算法的框架,自动搜索最适合脑电分类任务的Transformer结构。

相较于单模态信号,多模态信号能够从不方面反映人的情绪,能够更全面、客观和准确地对人的情绪进行表征。因此,基于多模态信息融合的情绪识别研究得到了学术界越来越多的关注(Zheng等, 2019)。构建多模态情感BCI的关键问题是如何有效地融合多模态信号,从而最大限度地利用这些信号中的信息。Liu等人(2022c)提出用加权融合和基于注意力融合的方法来扩展深度典型相关分析模型,使融合后的多模态特征在情绪上更具有同质性和辨别性,从而提升多模态情绪识别模型的性能和鲁棒性。Zhang等人(2021b)提出一种分层融合卷积神经网络模型,通过构造不同的网络层次结构,提取多尺度特征,并将网络特征与人工统计特征进行特征级融合,形成最终的特征向量,挖掘了多模态数据中的潜在信息。Saffaryazdi等人(2022)利用面部微表情、脑电、皮肤电及光体积描记信号的组合,在特征层面融合皮肤电和光体积描记信号,作为生

理特征,在策略层面融合面部微表情、脑电和生理特征,从而更可靠地检测潜在的情绪。

当被试的脑电信号或其他生理信号之间存在较大的个体差异时,直接使用传统的分类器通常难以构建准确的跨被试情绪识别模型(Morioka等, 2015)。对于个体差异问题,目前主流的解决方案是通过迁移学习构建更通用的情绪识别模型(Lin和Jung, 2017; Wu等, 2022)。脑电信号的非平稳性以及不同采集环境等差异导致每个被试的脑电数据分布不同,称为域偏移问题。迁移学习的核心思想是减少源域和目标域的域差异,使在源域上训练的模型可以泛化到目标域。其中比较经典的算法包括迁移成分分析(transfer component analysis, TCA)算法和直推式参数迁移(transductive parameter transfer, TPT)等方法(Zheng和Lu, 2016)。随着深度学习的发展,基于深度网络的迁移学习算法逐渐应用于跨被试情感BCI。Li等人(2018)和Luo等人(2018)提出了用于脑电情绪解码的域对抗神经网络和生成对抗域适应网络等。这些网络一般包含域判别器,判断样本是来自于源域还是目标域,在训练过程中通过反转域判别器分类损失的梯度,起到拉近源域和目标域边缘分布的作用。Wang等人(2021)提出深度多源自适应迁移网络(deep multi-source adaptation transfer network, DMATN),通过拉近源域和目标域数据在模型特定层的表征来实现领域自适应。

与单源域单目标域算法相比,多源域适应算法假设不同源域具有不同的边缘分布。在情感BCI问题中,这个假设比单源域假设更合理。不同受试者的脑电数据具有不同的边缘分布,但在单源域模型中,一般假设源域中的所有受试者属于同一个分布,而目标域是另一个分布。通过多源域适应算法,可以有效解决源域内不同主体边缘分布差异较大的问题,并在一定程度上降低负迁移的可能性。Li等人(2020a)使用风格迁移映射(style transfer mapping, STM),实现了多个源域的迁移学习。Cimtay和Ekmekcioglu(2020)利用在多被试脑电原始数据上预先训练好的Inception ResNet模型,实现了较高的跨被试分类性能。

上述迁移学习方法虽然不需要目标域样本的标签,但需要目标域的特征数据,称为域适应算法。迁移学习的另一个重要分支是域泛化算法,与域适应不同,域泛化算法不需要目标域中的数据,主要优点

是可以预测未知的目标域数据并具有很强的泛化性能(Ma等, 2019)。Shen等人(2022)通过对比学习(contrastive learning)算法,使训练集内不同被试对同一刺激的脑电信号表征尽可能相似,以此实现完全的域泛化模型。由于域泛化算法不利用任何目标域信息进行学习,因此性能通常低于域适应算法。Zhao等人(2021a)结合域泛化和域适应的思想,通过利用少量目标域数据而不是所有目标域特征数据来实现校准时间和模型性能之间的权衡。

多模态情感识别相较于单模态,能显著提高性能,但在实际场景中却受限于多模态信号采集的高成本和不便性。为了在不影响性能的前提下提高情感解码的泛化能力,跨模态情绪识别应运而生。脑电可与眼动等生理信号或图像等蕴含情感信息的外界信号共同使用,以训练跨模态模型。Jiang等人(2019)提出一种从眼动信号生成多模态特征的方法,首先建立一个双模态深度自编码器,将脑电和眼动作为输入,并转化为多模态表征,然后训练一个深度回归神经网络将眼动特征映射到多模态特征。考虑到从眼动到多模态特征的映射可能不是一个简单的回归,Yan等人(2021)采用条件生成对抗网络改进了这种方法,发现由条件生成对抗网络生成的多模态特征在不同情绪中更容易区分,因此取得了更好的性能。另外,Zhang等人(2022b)使用知识蒸馏方法将图像中的情绪知识迁移到基于脑电的情绪识别网络中,取得了良好效果,如图8所示。

考虑到脑电数据收集成本较高,一些研究人员专注于研究情感BCI的数据增强技术,即从现有的

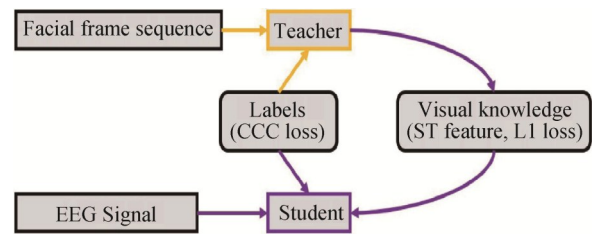


图8 利用知识蒸馏将情绪信息从图像迁移至脑电
(Zhang等, 2022b)

Fig. 8 Visual-to-EEG cross-modal knowledge distillation for continuous emotion recognition (Zhang et al., 2022b)

数据集中生成新的训练样本,使模型对实际应用中可能遇到的数据噪声具有鲁棒性。此外,增加训练集的规模有利于训练具有许多参数的更复杂的模型,并减少过拟合。Lashgari等人(2020)探究了噪声添加、生成对抗网络、滑动窗口、傅里叶变换、分割重组等方法在脑电数据增强中的应用,发现噪声添加和生成对抗网络比其他数据增强技术更能提高情感BCI的分类精度。近年来,基于深度生成学习的数据增强方法引起广泛关注,包括基于生成对抗网络和基于变分自编码器的方法。Luo等人(2020)使用变分自编码器和生成对抗网络,以功率谱密度和微分熵两种形式生成类似真实的脑电数据,如图9所示,并选择高质量的生成数据添加到训练数据集中,显著提高了基于脑电的情绪识别模型的性能。Bethge等人(2022)利用条件变分自编码器,学习与被试和情绪相关的脑电数据表示,生成的脑电序列与真实脑电信号数据相似,尤其能重构低频信号成分。

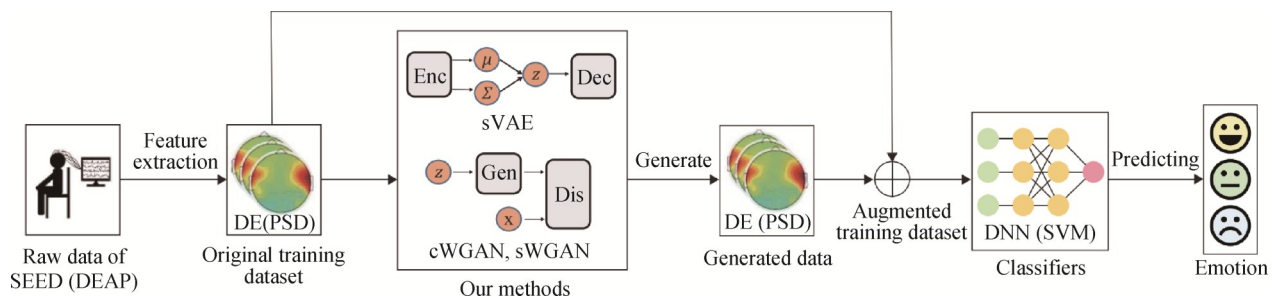


图9 使用变分自编码器和生成对抗网络增强脑电数据(Luo等, 2020)

Fig. 9 EEG data augmentation with variational autoencoder and generative adversarial network (Luo et al., 2020)

4.3 系统及应用研究进展

在医疗领域,随着精神疾病的患病率持续升高,迫切需要开发新的诊疗方法。由于多模态情感BCI技术可以实现对人的情绪进行客观评估,因此情感

BCI技术被认为是一种非常有发展前景的、可应用于情感障碍疾病诊疗的新技术。目前,情感BCI在医疗方面的应用处于快速研发阶段。美国谷歌公司Alphabet实验性研发Amber项目,旨在通过BCI设备

获取并分析脑电信号,开发针对抑郁和焦虑的客观测量方法,帮助医疗保健专业人员更容易、更客观地诊断抑郁症。上海瑞金医院成立脑机接口及神经调控中心,启动临床BCI研究项目“难治性抑郁症脑机接口神经调控治疗临床研究”,旨在通过多模态情感BCI和脑深部电刺激方法治疗难治性抑郁症。上海交通大学仿脑计算与机器智能研究中心与米哈游公司联合成立零唯一思公司,致力于开发情绪“X”机,一种基于多模态情感脑机接口的抑郁症客观评估系统。清华大学心理学系团队设计并实现了国际上首个基于脑电情绪响应的人格量化测评方法,并与博睿康公司合作,已经初步形成面向个体心理健康测评的软硬件一体化应用系统,并在北京、青岛等地展开了应用(Li等,2022c)。

在可穿戴设备应用上,出现了许多有关情感BCI的新兴产品。日本Neurowear公司开发的耳机可以根据佩戴者的心情选择音乐,生产的可穿戴相机可以检测用户的情绪,并在用户情绪投入时自动抓拍。加拿大InteraXon公司研发了脑波检测头环Muse,通过实时音频反馈帮助用户冥想、放松。HTC公司基于“情感云”推出心理评估与调节脑电虚拟现实设备,配备到国内700多所中小学的心理教室,实现对孩子的心理进行评估和干预。强脑科技、回车科技、脑陆科技等公司的可穿戴脑电产品,主要以智能头环为主,以脑电为基础,结合心率、皮肤电导率等多维传感器,用于用户的认知和情绪训练,实现对人体健康的监测以及对专注度、疲劳度和情绪等状态的识别。

5 脑—机接口总结与发展展望

5.1 总结

近年来,随着认知神经科学、信息科学和工程技术的发展和融合,BCI进入了一个快速发展的新阶段,在范式、算法和系统等方面均取得了诸多突破性进展。在系统范式方面,诸多新编码方法相继涌现并日益成熟,SSVEP-BCI和编码调制视觉诱发电位BCI均已实现对百余目标进行编码。将其他生理信号加入到传统的BCI中,进而形成的混合BCI范式在BCI的性能和系统灵活性方面均显现出优势。在解码算法研究方面,机器学习、黎曼几何和深度学习等算法应用于BCI研究,取得了显著的性能提升,直

接表现是先前持久的受试训练时间大幅缩短。在系统应用方面,各种BCI的应用不断涌现,大幅拓展了BCI的应用范围。专为交流和运动康复设计的辅助型/康复型BCI已广泛应用于实验研究和临床试验,并验证了这些系统适用于肌萎缩侧索硬化症、脑卒中和脊髓损伤患者。

5.2 脑—机接口技术展望

虽然BCI研究取得了可喜的成果,但目前BCI仍处在发展的初级阶段,绝大多数系统仍处于实验室研究阶段。在推进BCI实用化进程中,仍面临诸多问题与挑战。首先,BCI实验范式友好性差。尽管已经开发了许多BCI范式,但还没有一种实验范式是完美的,比较成熟的3类无创技术(运动想象、稳态视觉诱发电位和事件相关电位)也并不令人满意。MI-BCI需要较长时间的训练,且对于动作控制比较慢;SSVEP-BCI提供了无创BCI系统的最高信息传输率,但长时间注视闪烁的屏幕会导致视觉疲劳。一些其他范式,包括ERP-BCI,在操作中都要要求使用者高度集中注意力,这也会造成使用者疲劳。在实验范式方面,需要开发具有高效编码能力且用户友好的系统范式。其次,系统通讯速率低。低通讯速率是影响BCI推广应用的主要障碍之一。目前无创BCI系统的信息传输率不到6 bit/s,离正常的语音通讯速率(约39 bit/s)相差甚远。考虑深度融合人类智能和人工智能,使人类智能和人工智能能够自适应和高效地学习和协作,开发更加优秀的解码算法有望大幅提高BCI系统的通讯速率。最后,现有BCI系统大多只在实验室进行演示,离实际应用还很远。从研究系统到日常生活可用的辅助和康复设备的转换需要神经科学、计算机科学和临床康复等多学科努力与配合。一方面,更全面地了解大脑的功能和工作机制有助于优化BCI系统设计;另一方面,超高速宽带无线信号传输(5 G及以上)、云平台的超强数据处理能力等技术有望提高BCI系统的便携性。同时,还需要提高系统的可靠性和可及性,使BCI成为残障人士和健康受试者日常生活中不可或缺的工具。

随着科技的进步和研究的深入,大脑与外部设备的联系越来越紧密,它们之间的信息交流已经从感觉和知觉发展至认知,未来将达到无缝连接和认知协作。人类智能与人工智能的深度融合,展现了BCI未来发展的新趋势。

5.3 脑—机接口产业发展

随着BCI技术的研究和发展,BCI产业规模跟随技术进步而增长。根据美国麦肯锡公司的预测,全球BCI医疗应用的潜在市场规模在2030—2040年有望达到400亿~1 450亿美元,其中严肃医疗应用潜在规模为150亿~850亿美元,消费医疗应用潜在规模为250亿~600亿美元(范科峰等,2021)。2019年全球BCI市场规模为11.8亿美元,预计2027年将达到36.9亿美元,预期年复合增长率为15.5%(2021—2027年)。据麦肯锡The Bio Revolution Report研究显示,未来10~20年,BCI产业在全球每年直接的经济规模可达700亿~2 000亿美元。另外,根据华兴医疗的预测,中国严肃医疗领域BCI潜在市场规模未来有望超过1 000亿元,其中脑卒中康复和难治性癫痫治疗是最大的市场。由于BCI技术潜在市场规模巨大,很多公司纷纷涌入BCI赛道。目前,全球提供BCI产品和业务的公司有200多家,主要集中在中国和美国。国外具有代表性的公司包括美国Neuralink、美国NeuroSky和美国BrainGate等,国内具有代表性的公司包括博睿康、脑陆科技、中电云脑和脑虎科技等。此外,很多公司,包括美国Meta公司在内,都进行了BCI技术的布局。相关公司的产品类型主要包括BCI芯片硬件、BCI软件系统、BCI方案和专业应用等。从产品的成熟度来看,BCI技术处于初级阶段,产品上市的制度和标准、产品的安全性和有效性的认证制度等均不完善。与AI辅助诊断、智能医疗机器人等领域相比,BCI产品通过相关机构认证的数量屈指可数。总体来看,BCI产业化目前仍处于探索阶段,实现产业化仍需较长周期。

致 谢 本文由中国图象图形学学会机器视觉专委会组织撰写,该专委会链接为<http://www.csig.org.cn/detail/2386>。

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